

A Knowledge-Driven AI Framework for Legal Understanding and Trust Assessment

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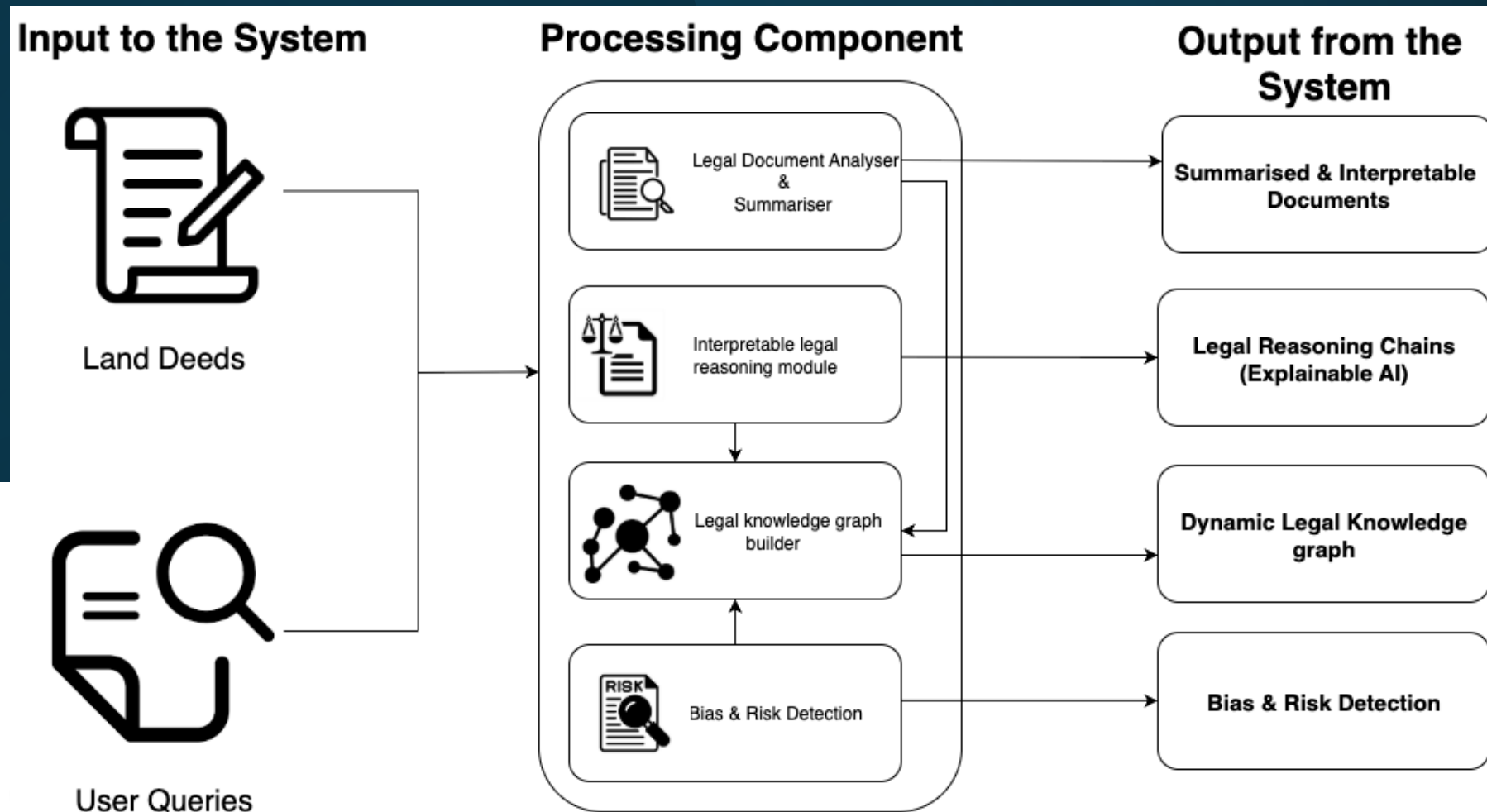
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A Knowledge-Driven AI Framework for Legal Understanding and Trust Assessment



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IT22113504 | WICRAMASINGHE D A T N



Visual-Integrated Multi Perspective Legal Summarizer

Information Technology Specialized in Data Science

VISUAL INTEGRATED MULTI PERSPECTIVE LEGAL SUMMARIZER

Solutions

Provides multi-perspective
summaries

1

Converts legal jargon
into plain language

2

Integrates visual
explanations

4

Ensures traceability

3

Combines text + visuals

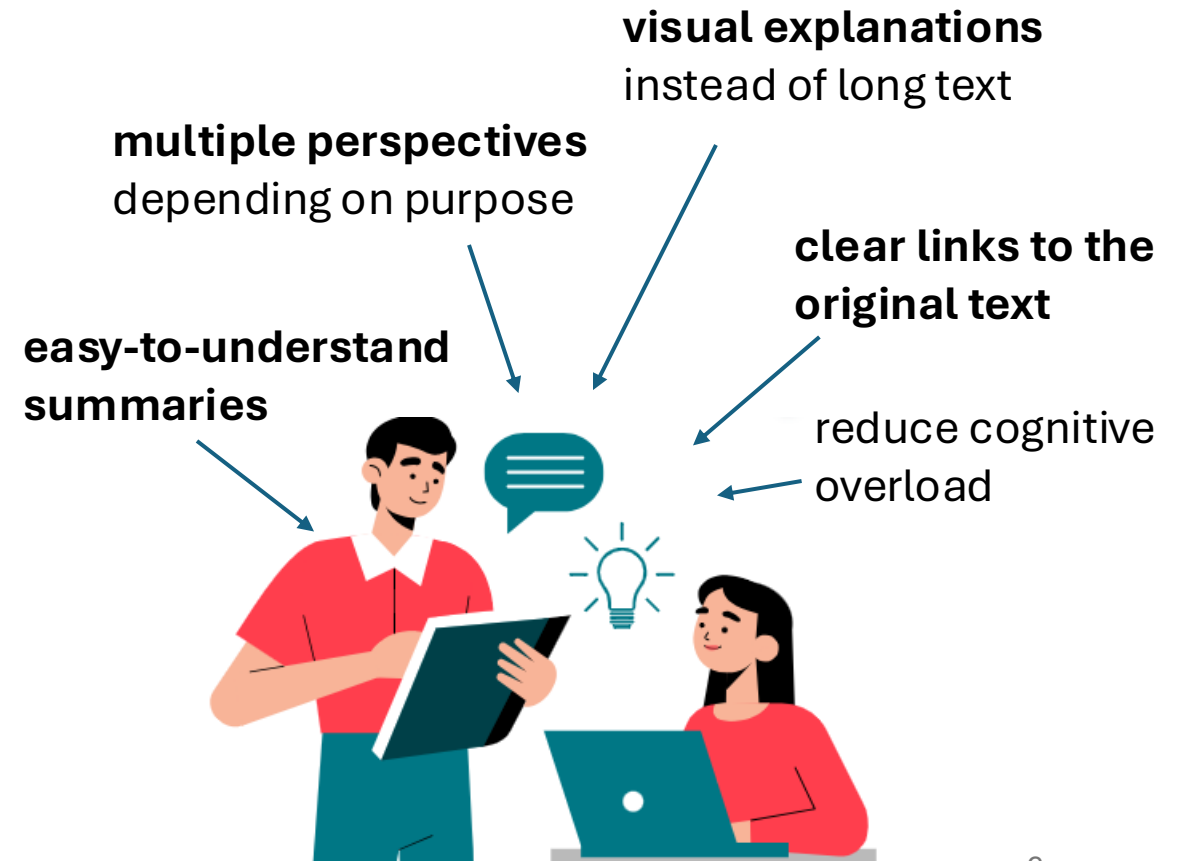
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Sub-problem:

Non-experts struggle to understand legal documents because existing systems lack multi-perspective summaries, plain-language explanations, traceability, and accurate visual representations.

User Requirements

Addressed by the Solution



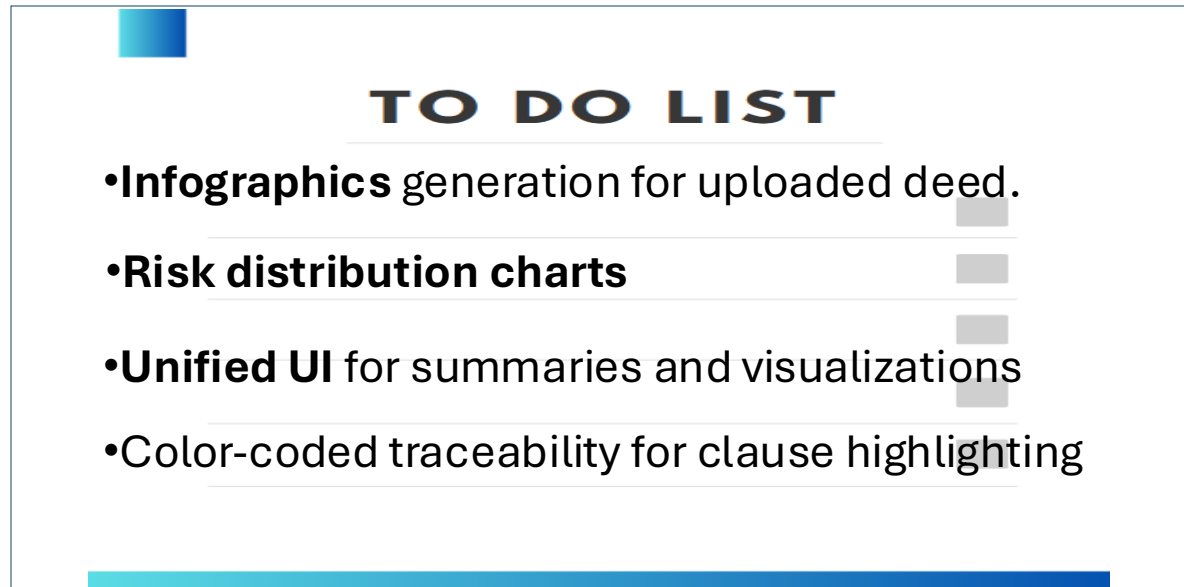
VISUAL INTEGRATED MULTI PERSPECTIVE LEGAL SUMMARIZER

Design Excellence / Contribution

- Designed a **perspective-aware summarization pipeline** instead of a single generic summary.
- **Integrated timelines visualization** modules to complement summaries.
- **Implemented the traceability**, linking summaries & visualizations directly to source text

User Feedback on Prototype

- **Summaries were clearer**
- **Users requested better risk highlighting**
- **Overall trust and understanding** improved
- **Timelines improved date identification**



IT22348784 | SIVANUJA S



EXPLAINABLE LEGAL REASONING MODULE

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EXPLAINABLE LEGAL REASONING MODULE

Solutions

Provides step-by-step legal reasoning instead of only final answers 1

Clearly explains how a legal conclusion is reached 2

Links each reasoning step to relevant laws and legal principles 3

Uses Chain-of-Thought and IRAC structure for clarity 4

Makes legal decisions transparent and traceable 5

Helps both legal experts and non-experts understand the reasoning 6

Improves trust and confidence in AI-generated legal outputs 7

User Requirements Addressed by the Solution

Step-by-step legal reasoning instead of only final answers

Traceability to statutes, ordinances, and legal principles

Clear explanation of how legal conclusions are derived

Understandable legal explanations for non-expert users

Increased trust and confidence in AI-assisted legal decisions



Sub-problem:

Users cannot fully trust AI-generated legal conclusions because existing systems act as **black boxes**. They provide final answers without explaining the legal reasoning, relevant laws, or decision steps, making validation and understanding **difficult for both experts and non-experts**.

EXPLAINABLE LEGAL REASONING MODULE

Design Excellence / Contribution

- Designed an **explainable legal reasoning pipeline** rather than a black-box AI model
- Implemented **Chain-of-Thought and IRAC frameworks** for structured legal reasoning
- Developed a **domain-specific dataset** for Sri Lankan property law
- Integrated legal references directly into reasoning outputs

User Feedback on Prototype

TO DO LIST

- Visual step-by-step legal reasoning flow
- Clause-level traceability and highlighting
- Risk indicators within reasoning steps
- Knowledge Graph integration for legal context

- Legal reasoning outputs were **clearer and easier to follow**
- Step-by-step explanations improved **understanding of legal decisions**
- Statute-based references increased **trustworthiness of outputs**
- Users requested **visual reasoning graphs** for complex cases
- Overall **confidence and transparency significantly improved**

IT22344892 | MAXWELL L Y

BIAS AND RISK EVALUATOR



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BIAS AND RISK EVALUATOR

Solutions

Utilizes NLP to automatically parse and segment complex legal sentences from property deeds.

1

Classifies individual clauses into "Bias" (Unfairness) or "Risk" (Danger) categories using trained models.

2

Assigns severity levels to detected risks to prioritize what the user needs to read first.

3

Provides a trust score to help users make decisions

4

Provides a prototype interface for instant text analysis and labeling

5

User Requirements Addressed by the Solution

Identifies if the agreement is one-sided

Gives the user assurance that no major red flags were missed.

Converts complex legal text into simple labels (Risk/Bias).

Automatically highlights dangerous clauses without needing a lawyer



Sub-problem:

Property/Land related deeds contain dense legal jargon where unfair terms ("Bias") and dangerous liabilities ("Risk") are often hidden. Manual review is time-consuming and error-prone for non-experts. Users struggle to distinguish between a "standard clause" and a "risky loophole," leading to potential financial loss or legal entrapment.

BIAS AND RISK EVALUATOR

Design Excellence / Contribution

- This separates "Subjective Bias" (Unfairness) from "Objective Risk" (Legal Liability).
- Model fine-tuned specifically on property/land deed datasets rather than general legal texts.
- Establishes the mathematical weightings required for the upcoming Trust Score calculation.

User Feedback on Prototype

TO DO LIST

- Implement Large Corpus Handling
- Develop Trust Score Algorithm
- Integrate Document Summarization
- Final UI/UX Polish

- Accuracy Validation:** Users confirmed that the model correctly identifies high-risk clauses in isolated tests.
- Granularity:** Feedback indicated that clause-by-clause analysis is helpful, but users lose track in long documents.
- Demand for Quantification:** Users requested a single "Safety Score" to summarize the document (validating the need for the future "Trust Score").

IT22732736 | SHARAN K



LEGAL KNOWLEDGE GRAPH BUILDER

Information Technology Specialized in Data Science

LEGAL KNOWLEDGE GRAPH BUILDER

Solutions

Extracts key legal entities (owners, land, deeds, locations)

1

Represents information as a Knowledge Graph instead of plain text

2

Connects entities using legal relationships (ownership, transfer, location)

3

Enables ownership-chain tracking with source-linked evidence

4

User Requirements Addressed by the Solution

Traceable answers linked to original deeds

Easy understanding of ownership history

Reduced dependency on complex legal language

Clear visualization of legal relationships



Sub-problem:

Property law documents contain complex, interconnected information such as ownership history, parties, land parcels, plans, and administrative regions. Existing systems treat these details as isolated text, making it difficult for users to understand **relationships**, **ownership chains**, and **contextual links** within and across deeds. This lack of structure reduces clarity, traceability, and trust in legal understanding.

LEGAL KNOWLEDGE GRAPH BUILDER

Design Excellence / Contribution

- Custom Knowledge Graph schema for Sri Lankan property law
- Graph-based modeling (Neo4j) for legal reasoning
- GraphRAG integration for explainable AI outputs
- Scalable design for future legal documents

TO DO LIST

- Extract key legal entities from property deeds ☐
- Build and populate the Knowledge Graph with relationships ☐
- Enable ownership-chain tracking and traceable links ☐
- Improve graph visualization and user understanding ☐

User Feedback on Prototype

- Ownership relationships became easier to understand
- Graph view was clearer than text-only explanations
- Traceability increased trust and confidence
- Users requested better visuals and risk indicators

A Knowledge-Driven AI Framework for Legal Understanding and Trust Assessment



Sri Lanka

WHO WE ARE

WHAT WE DO

OUR IMPACT

GET INVOLVED



[HOME](#) / [SRI LANKA](#) / OVER A THOUSAND VISITORS GAIN ACCESS TO LEGAL AID THROUGH THE 'KNOW YOUR NEETHI' LEGAL AID & AWARENESS CAMP IN BADULLA

Over a thousand visitors gain access to legal aid through the 'Know Your Neethi' Legal Aid & Awareness Camp in Badulla

The Support to Justice Sector (JURE) Project is embarking on the first of a series of legal aid and awareness camps designed to improve legal literacy and access to justice.

AUGUST 2, 2025



Badulla, Sri Lanka, 02 August 2025: The 'Know Your Neethi' Legal Aid & Awareness Camp held on 1st and 2nd August 2025 at the Badulla Cricket Stadium, successfully brought together over a thousand community members for two days of legal education, support, and empowerment.

The programme was carried out by the **Support to Justice Sector Project (JURE)**, funded by the European Union, jointly implemented by the United Nations Development Programme (UNDP) and the United Nations Children's Fund (UNICEF) Sri Lanka, and facilitated by the Ministry of Justice.

The legal aid and awareness camp served as a dynamic, community-focused platform to promote legal literacy and access to justice. This initiative placed a strong emphasis on inclusivity and equity, actively engaging traditionally underserved populations, including women, persons with disabilities, ethnic minorities, and other vulnerable and marginalized groups, ensuring their voices were heard and their rights addressed.

The event was also attended by key dignitaries including Hon. Harshana Nanayakkara, Minister of Justice; Ms. Ayesha Jinasena, President's Counsel - Secretary to the Ministry of Justice; Hon. Governor J.M. Kapila Jayasekera, Governor of Uva Province; Hon. Nandana Hapugoda, Mayor of Badulla; Ms. Azusa Kubota, Resident Representative, UNDP in Sri Lanka and Senior Professor Kolitha Bandara Wijesekara, Vice Chancellor – Uva Wellassa University.

Speaking about the event, **Hon. Harshana Nanayakkara, Minister of Justice** stated, *"This camp is a testament to our commitment to ensuring that the law serves everyone in Sri Lanka – not just in theory, but in practice. By bringing legal education and services to communities in Badulla, we are taking concrete steps to dismantle barriers and build trust in our justice system."*

Over the two days, attendees had access to a variety of engaging and informative activities designed in local languages to demystify the legal system and connect people with relevant services. These included interactive activities and games to explain legal principles and processes; information booths managed by government agencies, offering guidance on legal entitlements and public assistance programmes; and legal aid clinics where individuals received free, one-on-one consultations with legal professionals on pressing issues such as land disputes, employment rights, language rights, family law, and cybercrime. Forum theatre performances brought everyday legal challenges to life through dramatized storytelling, making complex legal issues more understandable and relatable, while psychosocial support services were made available to assist individuals experiencing emotional distress related to their legal circumstances.

A Knowledge-Driven AI Framework for Legal Understanding and Trust Assessment

Customer Persona

- **Property buyers and owners**
 - Want to understand legal documents easily
- **Legal professionals**
 - Lawyers and legal assistants
- **General public**
 - People with little legal knowledge

Market Size

- High number of **property and land transactions** in Sri Lanka
- **Property buyers and owners** must review legal deeds and agreements
- Majority of citizens **lack legal knowledge** in property law
- **Lawyers and notaries** review property documents daily
- No simple **AI-based solution focused only on Sri Lankan property law**
- Strong potential demand from:
 - Property buyers
 - Property owners
 - Legal professionals

What Is Unique in This Solution

- **Focuses only on Sri Lankan property law**
- Explains property documents in **simple language**
- Shows **step-by-step legal explanations**
- **Highlights legal risks** and unclear clauses
- Links clauses to **relevant Sri Lankan property laws**
- Uses **visual summaries and diagrams**

How Will the Cost Be Recovered

- **Paid subscriptions**
 - For detailed analysis and reports
- **Licensing**
 - Law firms and property-related organizations
- **Free public version**
 - Basic explanations for citizens



THANK YOU!!